Video Object Tracking on MOT16

Faster R-CNN detection with Siamese re-identification

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Midterm Presentation

Group 4
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Team

Ben Blake, detection and tracking
Faster R-CNN, data association, integration

Tina Nguyen, re-identification
Siamese network, contrastive loss

Geethika Padamati, data pipeline
MOT16 parsing, datasets, augmentation

Sanjana Aileni, evaluation and demo
Metrics, visualization, result video

Problem

Given a video, **produce** a box around each pedestrian in every frame, with the same identity number following the same person.

Detection is not enough

A detector answers “where are the people” separately in each frame. It has no way of knowing that the person in frame 10 is the person it saw in frame 9.

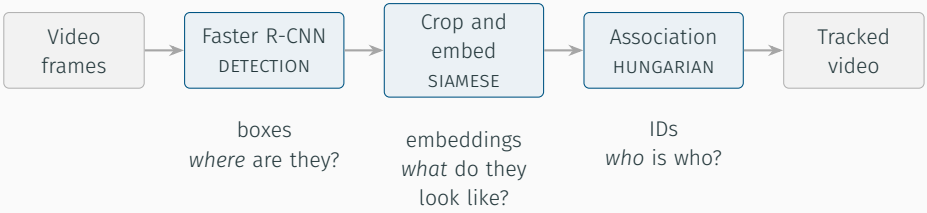
Tracking adds identity

We have to link detections across frames, through occlusion, through people crossing, and through frames where the detector misses someone entirely.

Why identity, and not just location

Driving, surveillance, sports and robotics all need to know *which* person, and for how long. Our target: track one pedestrian across an MOT16 sequence without losing them (rubric, grade A).

System Architecture



Offline tracking. We process a batch of frames rather than streaming them, which lets us recover from short occlusions.

Tracking by detection. Detection and identity stay separate, so either component can be replaced without touching the other.

Detection: Faster R-CNN

Why this model

- The Region Proposal Network generates proposals inside the network, so no Selective Search
- Proposals and detection share one backbone, so each image is convolved once
- Good accuracy on small objects, and distant pedestrians are small

Few-shot fine-tuning

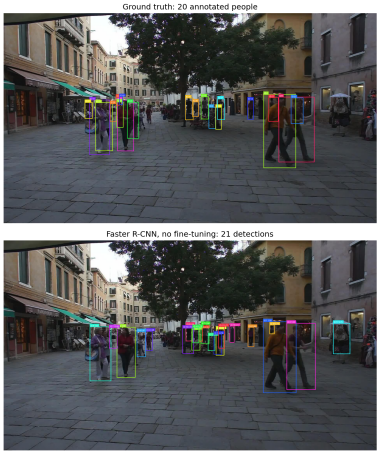
We freeze the ResNet50-FPN backbone and train only the detection heads, as the assignment suggests.

Parameter budget

Total	41.3 M
Frozen	26.8 M
Trainable	**14.5 M**

Training 35% of the parameters keeps the job inside Colab's memory budget.

What the detector actually sees



Frame 1 of MOT16-02: ground truth above, our detections below. The detector finds the foreground and misses the small, distant and partly hidden people. That is where the missing 60% of recall

Association Problem

Each frame gives N detections and we are holding M tracks. Which detection continues which track?

Overlap alone fails

When two people cross, each box overlaps the other's previous position, so position cannot separate them.

Appearance alone fails

Two people in similar clothing produce similar embeddings, and appearance drifts as someone turns.

Blended cost

$$C_{ij} = (1-\lambda)\underbrace{(1-\mathrm{IoU}_{ij})}_{\text{position}} + \lambda\underbrace{(1-\cos(e_i, e_j))}_{\text{appearance}}$$

Minimised with the Hungarian algorithm, giving a globally optimal one-to-one assignment.

Pairs failing the IoU or appearance gate are rejected before matching. A track survives `max_age` unmatched frames, so a brief occlusion does not create a new identity.

The MOT16 Dataset

- 14 sequences: 7 training with ground truth, 7 test
- Crowded streets, both moving and static cameras
- Heavy occlusion, with people leaving and re-entering the frame

One annotation row

`1, 3, 586, 447, 85, 263, 1, 1, 1`

Columns	
frame	1
id	3
bb_left	586
bb_top	447
bb_width	85
bb_height	263
conf	1 (0 means ignore)
class	1 (pedestrian)
visibility	1.0

Two traps in the annotations

Coordinates are 1-indexed

The files count pixels from 1 and NumPy counts from 0. If you miss this, every box is off by one pixel. The output still looks correct, but IoU drops and the metrics get worse for no visible reason.

Not every person is a target

Classes 7, 8 and 12 (static person, distractor, reflection) are people but are not scored. Training on them teaches the detector to fire on mannequins and reflections.

Both are handled in `mot/data/mot16.py` and covered by unit tests. We wrote the tests first because neither failure produces an obvious symptom.

Data Augmentation

MOT16 gives us few scenes, so the model can memorise those scenes instead of learning what a pedestrian looks like.

Detection

- Horizontal flip, boxes mirrored to match
- Colour jitter: brightness, contrast, saturation

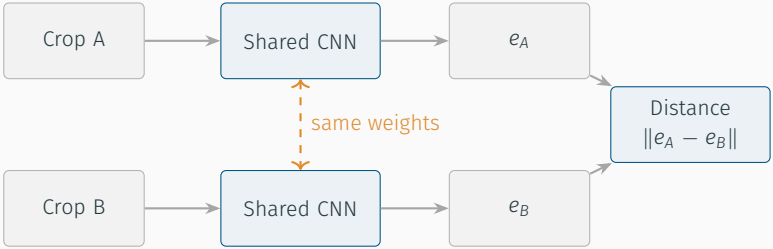
Re-ID crops

- Random crop and horizontal flip
- Random erasing, simulating occlusion

What we leave out

No vertical flips and no large rotations. Pedestrians are upright in every frame of every sequence, so an upside-down person is a pose the model will never see at test time.

Re-identification: a similarity model, not a classifier



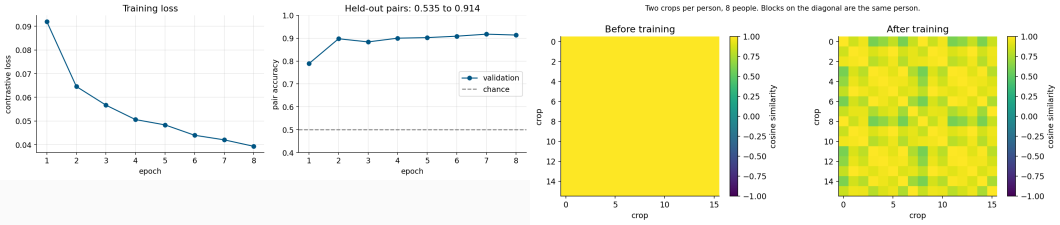
Four convolution blocks, adaptive pooling, then a 128-dimensional L2-normalised embedding. 1.5 M parameters, so it trains in minutes.

Contrastive loss

$$L = \tfrac{1}{2}\left[y\,d^2 + (1-y)\max(0, m-d)^2\right]$$

Matching pairs are pulled together, non-matching pairs pushed past the margin m.

Training the similarity model



Eight epochs on crops cut from MOT16 ground truth. Pair accuracy rises from 0.535, which is chance, to 0.932 on held-out pairs. The matrices show the same crops before and after: the blocks on the diagonal are the network learning who is who.

A problem we had to resolve

The course slides give four different crop sizes

Training is specified at 16×16, inference at 24×24, retrieved crops at 30×16, and the reference *`Siamese_Network`* code uses *`nn.Linear(128*22*22, ...)`*, needing roughly 102×102. No single size satisfies all four.

What we chose

- 128×64, the resolution Market-1501 normalises its own images to
- *`AdaptiveAvgPool2d`* instead of a hardcoded flatten

Why

- 16×16 cannot resolve clothing, which is the signal re-identification relies on
- Adaptive pooling accepts any input size, so the choice is not baked into the architecture

How we measure success

CLEAR-MOT metrics

$$\mathrm{MOTA} = 1 - \frac{\mathrm{FN} + \mathrm{FP} + \mathrm{IDSW}}{\text{ground truth boxes}}$$

- **FN:** a person we failed to detect
- **FP:** a box with nobody in it
- **IDSW:** the ID changed on one person

MOTP is the mean IoU of matched boxes, so it measures how tightly the boxes fit.

Why ID switches matter

A missed frame can be recovered on the next one. An ID switch means we have confused two people, which is the failure the re-identification component exists to prevent and the one the rubric grades.

Results: first end-to-end run

MOT16-02, 600 frames, on a Colab T4. COCO-pretrained detector, **no fine-tuning yet**; re-identification trained for 8 epochs on MOT16 crops.

Against the rubric

One real pedestrian is held by a single id for **218 of the 230 frames they appear in (94.8%)**, before any fine-tuning.

But the **median across 44 people is 22.4%**. One person is tracked well; most are not.

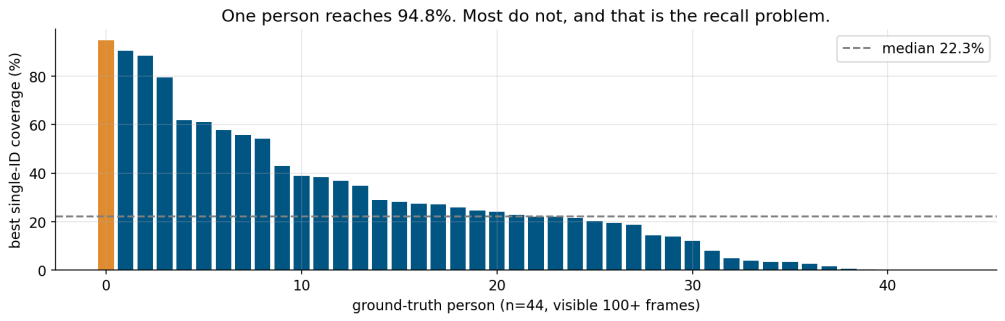
Measured against the ground-truth person, not whichever track lived longest. Re-ID

Whole sequence

MOTA	0.232
MOTP	0.792
True positives	7,004
False positives	2,713
False negatives	10,829
ID switches	151

MOTP 0.792 means the boxes are tight. MOTA is held down almost entirely by false negatives.

Coverage across every person



Best single-ID coverage for each of the 44 people visible for 100 frames or more. One person is tracked almost perfectly; the median is 22.4%. Closing that gap is a detection problem, not a tracking one.

Does re-identification help? Not yet, and we know why

Ablation, 600 frames

	with	without
MOTA	0.232	0.232
MOTP	0.792	0.792
IDSW	151	148

Appearance changes nothing.

Detection threshold sweep, 150 frames

thresh	MOTA	recall	FN	IDSW
0.50	−0.006	0.509	1,676	80
0.60	0.105	0.490	1,741	72
0.70	0.190	0.455	1,860	34
0.80	**0.268**	0.400	2,049	15

What this tells us

False negatives outnumber ID switches seventy to one, and lowering the threshold lifts recall only 0.40 to 0.51 while false positives quadruple. Recall is capped by people the COCO model never finds: small, distant, occluded. **Detection is the bottleneck, so association has nothing left to disambiguate.** That is the case for fine-tuning, and it is what we do next.

Demo



demo_target_clip.mp4, 8 seconds. Target in colour, every other track in grey.

Timeline to the final

Phase	Work	Owner
Done	Pipeline, parsing, re-ID, metrics, CI	All
Done	Baseline run, ablation, demo video	Sanjana
Next	Detector fine-tuning (recall is capped)	Ben
Next	Re-ID training on real data	Tina
Next	Augmentation tuning	Geethika
Then	Threshold sweep and ablations	Sanjana, Ben
Final	Report, final video, presentation	All

Code: *github.com/ben-blake/cs5567-project*